DATA SCIENCE LABORATORY MANUAL

Module III ? Data Visualization

Topic: Exploratory Data Analysis

OVERVIEW & LEARNING OBJECTIVES

Univariate, bivariate, and multivariate visualization techniques to uncover patterns and anomalies.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Exploratory Data Analysis in modern Data Science workflows.
- Detailed breakdown: Univariate, bivariate, and multivariate visualization techniques to uncover patterns and anomalies.
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.